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SAT · EXAM PREP

SAT Math: Algebra & Advanced Math

– one exam, every section, every strategy

9 Sections

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Digital SAT

13 sections

SAT Math: Algebra & Advanced Math is the foundation of the Digital SAT Math section: together these two content domains are about 70% of the scored Math questions (roughly 13 to 15 Algebra and 13 to 15 Advanced Math out of 40), so they are the single biggest lever on your Math score.

- **How the Digital SAT Math section assesses this** Share of scored Math questions ~70% · Scored Math questions (total) 40 · Multiple choice 4 choices (A to D)
- **Key terms** Balance principle, Slope (m), Slope-intercept form, System of equations
- **How to study for SAT Math: Algebra & Advanced Math** Treat this as the foundation domain. It is about 70% of the scored Math questions, so the time you put here moves your score more than anywhere else.
- **SAT Math: Algebra & Advanced Math FAQ** How much of SAT Math is Algebra and Advanced Math?

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Unblock All

What SAT Math: Algebra & Advanced Math covers

The whole domain → one exam-ready map. Nine parts, core Algebra through Advanced Math, then 40 worked practice questions. Each links to its free chapter guide.

- | | |
|---|---|
| 01 Linear equations in one variable | Chapter 1. The balance principle, clearing parentheses and fractions, collecting like terms, expressions vs equations, and the special cases (no solution vs infinitely many solutions). |
| 02 Linear equations in two variables | Chapter 2. Slope as a constant rate of change, x- and y-intercepts, slope-intercept and point-slope forms, function notation $f(x)$, and reading a line off its graph. |
| 03 Systems of linear equations | Chapter 3. Substitution and elimination, the three solution cases (one, none, infinitely many), and the SAT shortcut of finding a combination such as $x + y$ without fully solving. |
| 04 Linear inequalities | Chapter 4. Solving, when to flip the sign (only on multiply or divide by a negative), graphing and shading, translating worded constraints, and greatest- or least-integer answers. |
| 05 Equivalent expressions | Chapter 5. Rewriting rather than solving: distribution and FOIL, factoring, combining and simplifying rational expressions, and exponent rules. |
| 06 Nonlinear equations | Chapter 6. Quadratics three ways (factoring with the zero-product rule, the quadratic formula, completing the square), the discriminant, and extraneous-solution checks. |
| 07 Nonlinear functions | Chapter 7. The three forms of a quadratic (standard, factored, vertex) and what each reveals, the vertex and axis of symmetry, and exponential growth and decay models. |
| 08 Formulas to memorize | Chapter 8. The algebra toolkit on one page: slope, the line forms, the quadratic formula and discriminant, the exponential model, and the exponent and factoring rules worth knowing cold. |
| 09 Practice: 40 questions with solutions | Chapter 9. Forty exam-difficulty questions across Algebra and Advanced Math, in multiple-choice and student-produced-response formats, each with a full worked solution and the trap behind every wrong choice. |

It spans the full arc from core Algebra (linear equations in one and two variables, linear functions, systems, and inequalities) into Advanced Math (equivalent expressions, nonlinear equations such as quadratics, and nonlinear functions). Each skill is taught from the ground up with fully worked, step-by-step examples and labelled graphs.

The Digital SAT presents these questions in two formats: multiple choice (four answer choices, A to D) and student-produced response (you type the answer directly; fractions or decimals are accepted, with no percent sign, dollar sign or commas).

This guide teaches the method behind each question type, then drills it with 40 exam-difficulty questions across Easy, Medium and Hard, each worked in full with the trap behind every wrong choice. The aim is a reliable process for each type: set to zero, factor or formula, then check.

How the Digital SAT Math section assesses this

Only what this guide can support from the question set is stated below. Confirm current section structure, timing and tools on the official College Board test specifications before you rely on them.

ITEM	WEIGHT / COUNT	WHAT IT MEANS
Share of scored Math questions	~70%	Algebra and Advanced Math together make up about 70% of the scored Math questions: roughly 13 to 15 Algebra and 13 to 15 Advanced Math out of 40. They are the single biggest lever on the Math score.
Scored Math questions (total)	40	This guide drills 40 exam-difficulty questions spanning the two domains, matching the count of scored Math questions.
Multiple choice	4 choices (A to D)	Each multiple-choice question gives four answer choices, A to D, with exactly one correct answer.
Student-produced response	Type the answer	You enter the answer directly. Fractions or decimals are accepted; no percent sign, dollar sign or commas. In this question set, 12 of the 40 are this format.
Difficulty	Easy / Medium / Hard	Questions are tagged by difficulty across all topics, so you can build from confidence questions up to the hardest Advanced Math.

WORKED EXAMPLE • FREE

Translate a word problem into a linear equation, then solve

Q. A gym charges a \$40 joining fee plus \$25 per month. After how many months m does a member pay \$215 in total? (A student-produced-response style question: solve for the value of m .)

- Step 1** Translate the words into an equation. Total = fee + rate $\times$ months, so $40 + 25m = 215$. The joining fee is a one-off constant; the \$25 is the per-month rate that multiplies the number of months.
- Step 2** Isolate the variable term using the balance principle. Subtract 40 from both sides: $25m = 175$.
- Step 3** Undo the multiplication. Divide both sides by 25: $m = 7$.
- Step 4** Check by substituting back: $40 + 25(7) = 40 + 175 = 215$. It matches, so $m = 7$ is correct.

Answer: $m = 7$ months. The member reaches \$215 in total after 7 months.

Sia tip — the move that wins linear word problems is translation, not arithmetic. Name what each number does (constant fee vs per-unit rate). write Total = constant + rate $\times$ variable. then solve and always substitute back. Watch the classic trap: read what is actually asked. If a question

wants an expression such as $6x$ minus 3 rather than x alone, solve for that expression directly when you can.

GLOSSARY

Key terms

Balance principle

Whatever you do to one side of an equation you must do to the other. It is the basis of solving every equation in this domain.

Slope (m)

The constant rate of change of a line: rise over run, equal to $(y_2 \text{ minus } y_1) \text{ over } (x_2 \text{ minus } x_1)$. Parallel lines have equal slopes; perpendicular lines have negative-reciprocal slopes.

Slope-intercept form

$y = mx + b$, the most graph-ready form of a line. Here m is the slope and b is the y -intercept (the value when $x = 0$, often the starting amount).

System of equations

Two or more equations in the same unknowns; the solution is the point or points satisfying all of them, which graphically is the intersection. It can have one solution, none (parallel lines), or infinitely many (the same line).

Equivalent expressions

Expressions that are equal for every value of the variable (a rewrite), as opposed to an equation you solve. Distribution, FOIL and factoring all produce equivalent expressions.

Discriminant

The quantity b squared minus $4ac$ inside the quadratic formula. Its sign gives the number of real solutions: two if positive, one if zero, none if negative.

Quadratic formula

$x = (\text{negative } b \text{ plus or minus the square root of } (b \text{ squared minus } 4ac)) \text{ over } 2a$. It solves any quadratic $ax^2 + bx + c = 0$.

Vertex

The turning point (h, k) of a parabola: its maximum if the parabola opens down, or its minimum if it opens up. The axis of symmetry is the vertical line $x = \text{negative } b \text{ over } 2a$, which passes through the vertex.

Exponential model

A quantity of the form a times b to the power t , or $a(1 \text{ plus or minus } r)$ to the power t . Here a is the initial amount (the y -intercept); a base greater than 1 grows and a base between 0 and 1 decays.

FAQ

SAT Math: Algebra & Advanced Math FAQ

► How much of SAT Math is Algebra and Advanced Math?

► [How do I solve a linear equation?](#)

► [What is slope–intercept form and how do I read it?](#)

► [When do I flip the inequality sign?](#)

► [What are the three ways to solve a quadratic?](#)

► [What do the three forms of a quadratic reveal?](#)

► [How do exponential models work on the SAT?](#)

RELATED • [THE REST OF SAT MATH](#)

Related SAT sections

This is one of the three SAT Math content domains. Work across all three, then use the exam total page to plan your full Math section.

01 SAT Math: Algebra & Advanced Math This guide. Linear equations, lines, systems, inequalities, equivalent expressions, quadratics and nonlinear functions — about 70% of the scored Math questions.

02 SAT Math: Problem–Solving & Data Analysis Ratios, rates and proportions, percentages, units, and reading data from tables and graphs.

03 SAT Math: Geometry & Trigonometry Lines and angles, triangles, circles, area and volume, and right–triangle trigonometry.

→ **SAT exam overview** The full Digital SAT: how the Math and Reading and Writing sections fit together, scoring, and timing.

STUDY STRATEGY

How to study for SAT Math: Algebra & Advanced Math

Treat this as the foundation domain. It is about 70% of the scored Math questions, so the time you put here moves your score more than anywhere else. (1) Work it in three passes: learn the method from each worked example (cover the solution and try the steps yourself first), drill the practice questions cold, then re–do the examples until the process is reflex.

(2) Build a reliable process for each question type rather than memorising answers: for an equation, set to zero, factor or use the formula, then check; for a line, read slope and intercept off $y = mx + b$; for a system, choose substitution or elimination, and look for a combination such as $x + y$ you can reach without fully solving.

(3) Name the traps before they catch you: distributing to only the first term, sign errors when

moving a term across the equals sign, dividing only part of a side, flipping the inequality only when you multiply or divide by a negative, and answering x when the question asked for an expression.

(4) Master the term-vs-term distinctions the test rewards: expression vs equation, slope-intercept vs point-slope form, the three solution cases of a system, the three forms of a quadratic, and growth vs decay. (5) Practise both formats: multiple choice (four choices) and student-produced response (type the answer; fractions or decimals, no percent sign, dollar sign or commas).

(6) Confirm the current section structure, timing and on-screen tools on the official College Board specifications, since this guide states only what its question set supports.

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